

04_profiles — naming the segments

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1. Goal

Take the $k = 3$ cluster labels from 03_clustering.Rmd and turn them into named, interpretable customer personas. Output: ../data/customer_features_labeled.csv.

2. Load + validate

```
assignments <- read_csv("../data/processed/customer_clusters.csv",
                        show_col_types = FALSE)
customer     <- read_csv("../data/processed/customer_features.csv",
                        show_col_types = FALSE)
tx           <- read_csv("../data/interim/online_retail_clean.csv",
                        show_col_types = FALSE)
cancelled    <- read_csv("../data/interim/online_retail_cancelled.csv",
                        show_col_types = FALSE)

tx$InvoiceDate <- as_datetime(tx$InvoiceDate)
cancelled$InvoiceDate <- as_datetime(cancelled$InvoiceDate)
tx <- tx |>
  mutate(Description = iconv(Description, from = "", to = "UTF-8", sub = ""))

stopifnot(
  "CustomerID" %in% names(assignments),
  "CustomerID" %in% names(customer),
  "CustomerID" %in% names(tx),
  !anyDuplicated(customer$CustomerID),
  setequal(assignments$CustomerID, customer$CustomerID)
)

cust <- customer |>
  inner_join(assignments |> select(CustomerID, cluster),
            by = "CustomerID") |>
  mutate(cluster = factor(cluster))

list(customers_clustered = nrow(cust),
      k                   = n_distinct(cust$cluster),
      cluster_sizes      = table(cust$cluster))
```

```
## $customers_clustered
## [1] 4334
##
## $k
## [1] 3
##
## $cluster_sizes
##
##      A      B      C
## 1396 1804 1134
```

2a. Re-label clusters in stable $A \rightarrow B \rightarrow C$ order

K-means cluster IDs depend on the random seed. Re-label by descending median `log_NetMonetary` so cluster A is always the highest-value group, C the lowest, regardless of the seed.

```
order_by_value <- cust |>
  group_by(cluster) |>
  summarise(med = median(log_NetMonetary), .groups = "drop") |>
  arrange(desc(med)) |>
  mutate(label = LETTERS[seq_len(n())]) |>
  select(cluster, label)

cust <- cust |>
  left_join(order_by_value, by = "cluster") |>
  mutate(cluster = factor(label, levels = LETTERS[seq_along(unique(label))])) |>
  select(-label)

tx <- tx |>
  inner_join(cust |> select(CustomerID, cluster), by = "CustomerID")

table(cust$cluster)

##
##      A      B      C
## 1396 1804 1134
```

3. Headline summary — share of customers vs share of revenue

```
total_rev_net <- sum(cust$NetMonetary)

headline <- cust |>
  group_by(cluster) |>
  summarise(
    customers      = n(),
    revenue_net    = sum(NetMonetary),
    revenue_gross  = sum(GrossMonetary),
    median_net_spend = median(NetMonetary),
    median_orders  = median(Frequency),
    median_recency_d = median(Recency),
```

```

    .groups = "drop"
  ) |>
  mutate(pct_customers = customers / sum(customers),
         pct_revenue   = revenue_net / total_rev_net) |>
  arrange(cluster)

headline |>
  mutate(across(c(revenue_net, revenue_gross, median_net_spend),
                 \ (x) dollar(x, prefix = pound)),
         across(c(pct_customers, pct_revenue),
                 \ (x) percent(x, accuracy = 0.1))) |>
  kable(caption = "Cluster headlines - share of customers vs revenue (net)")

```

Table 1: Cluster headlines — share of customers vs revenue (net)

cluster	customers	revenue_net	revenue_gross	median_net_spend	median_order	median_revenue	pct_customers	pct_revenue
A	1396	£6,574,443	£6,734,530	£2,264.95	6	16.0	32.2%	79.5%
B	1804	£1,447,805	£1,741,381	£600.46	2	70.5	41.6%	17.5%
C	1134	£247,152	£261,317	£178.61	1	156.0	26.2%	3.0%

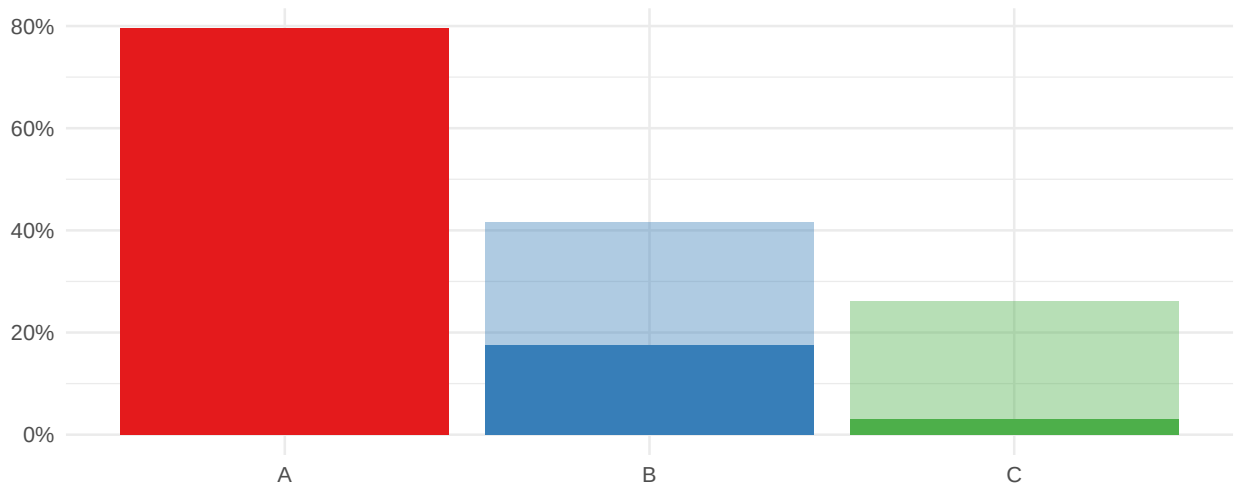
The most important number above is **share of revenue vs share of customers**. If cluster A holds ~28% of customers but ~75% of net revenue, that's the segment to protect; the inverse pattern in cluster C identifies the segment to either nurture or accept as low-margin.

```

ggplot(headline, aes(cluster, fill = cluster)) +
  geom_col(aes(y = pct_customers), alpha = 0.4, position = "identity") +
  geom_col(aes(y = pct_revenue), alpha = 1.0, position = "identity") +
  scale_y_continuous(labels = percent_format()) +
  scale_fill_brewer(palette = "Set1") +
  labs(title = "Customers (faded) vs net revenue (solid) by cluster",
       y = NULL, x = NULL) +
  theme(legend.position = "none")

```

Customers (faded) vs net revenue (solid) by cluster



4. Feature profile — median AND mean

Median is robust to outliers; mean is what aggregates revenue forecasts. We report both because the distributions are heavy-tailed.

```
profile <- cust |>
  group_by(cluster) |>
  summarise(
    n = n(),
    Recency_med = median(Recency),
    Recency_mean = mean(Recency),
    Frequency_med = median(Frequency),
    Frequency_mean = mean(Frequency),
    NetMonetary_med = median(NetMonetary),
    NetMonetary_mean = mean(NetMonetary),
    AvgOrderValue_med = median(AvgOrderValue),
    AvgOrderValue_mean = mean(AvgOrderValue),
    UniqueProducts_med = median(UniqueProducts),
    UniqueProducts_mean = mean(UniqueProducts),
    AvgUnitPrice_med = median(AvgUnitPrice_winsor),
    TenureDays_med = median(TenureDays),
    SinglePurchase_share = mean(SinglePurchase),
    ReturnRate_med = median(ReturnRate),
    ReturnRate_mean = mean(ReturnRate),
    ReturnValueRate_med = median(ReturnValueRate),
    ReturnValueRate_mean = mean(ReturnValueRate),
    .groups = "drop"
  )

profile |>
  mutate(across(c(NetMonetary_med, NetMonetary_mean,
    AvgOrderValue_med, AvgOrderValue_mean,
    AvgUnitPrice_med),
    \(x) dollar(x, prefix = pound, accuracy = 0.01)),
    across(c(SinglePurchase_share,
    ReturnRate_med, ReturnRate_mean,
    ReturnValueRate_med, ReturnValueRate_mean),
    \(x) percent(x, accuracy = 0.1))) |>
  kable(caption = "Median and mean profile per cluster (original scales)")
```

Table 2: Median and mean profile per cluster (original scales)

cluster	Recency_med	Recency_mean	Frequency_med	Frequency_mean	NetMonetary_med	NetMonetary_mean	AvgOrderValue_med	AvgOrderValue_mean	AvgUnitPrice_med	AvgUnitPrice_mean	SinglePurchase_share	ReturnRate_med	ReturnRate_mean	ReturnValueRate_med	ReturnValueRate_mean
A	1396.0	24.58739	9.34097	21.2649	15709.3	15.49450	98.5	125.1346	77.293	0.0%	12.5%	13.7%	0.5%	2.0%	
B	1804.0	102.0621	2.06818	300.48	802.53	30.98540	98.0	42.9939	0.6741	38.1%	0.0%	9.6%	0.0%	1.5%	
C	11345.0	161.66226	1.43827	278.61	217.95	145.91	172.170	0	12.32099	2.18	72.1%	0.0%	6.5%	0.0%	3.2%

5. Distributions by cluster

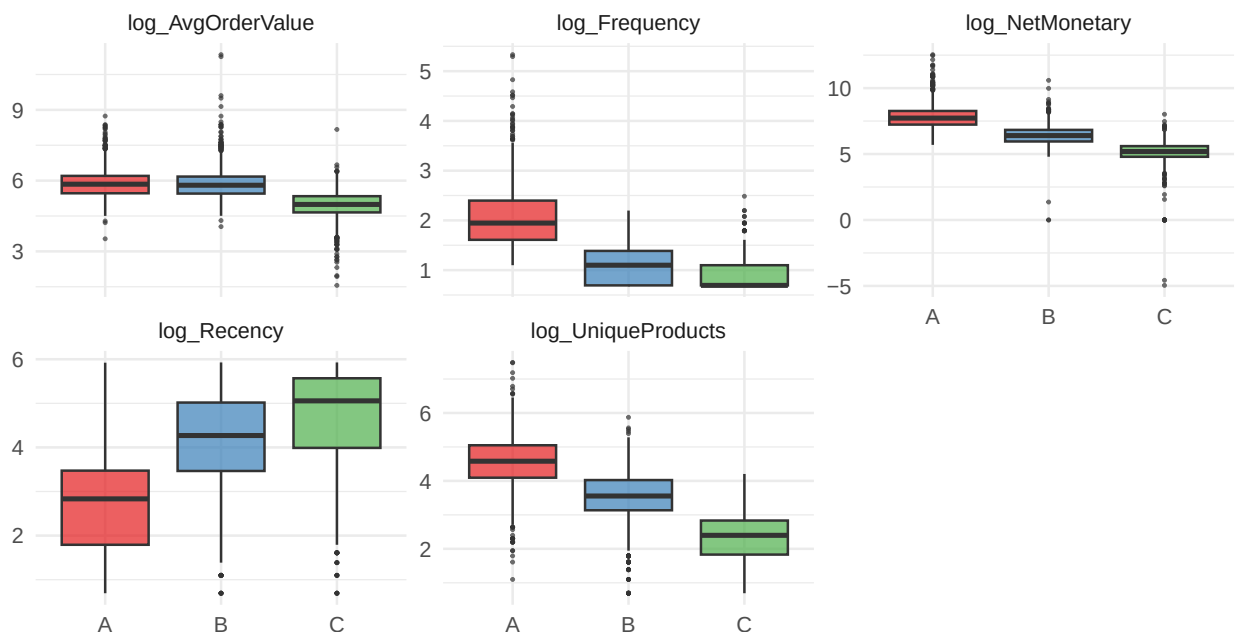
A median can mislead when distributions are skewed. Boxplots on the log scale show the spread within each cluster.

```

cust |>
  select(cluster, log_Recency, log_Frequency, log_NetMonetary,
          log_AvgOrderValue, log_UniqueProducts) |>
  pivot_longer(-cluster, names_to = "feature", values_to = "value") |>
  ggplot(aes(cluster, value, fill = cluster)) +
  geom_boxplot(alpha = 0.7, outlier.size = 0.4) +
  facet_wrap(~ feature, scales = "free_y") +
  scale_fill_brewer(palette = "Set1") +
  labs(title = "Feature distributions by cluster (log scale)",
       x = NULL, y = NULL) +
  theme(legend.position = "none")

```

Feature distributions by cluster (log scale)



6. Parallel-coordinate signature

A single picture summarising “what makes each cluster different from the average customer.” Each line is a cluster centroid in z-scored feature space — positive = above population mean, negative = below.

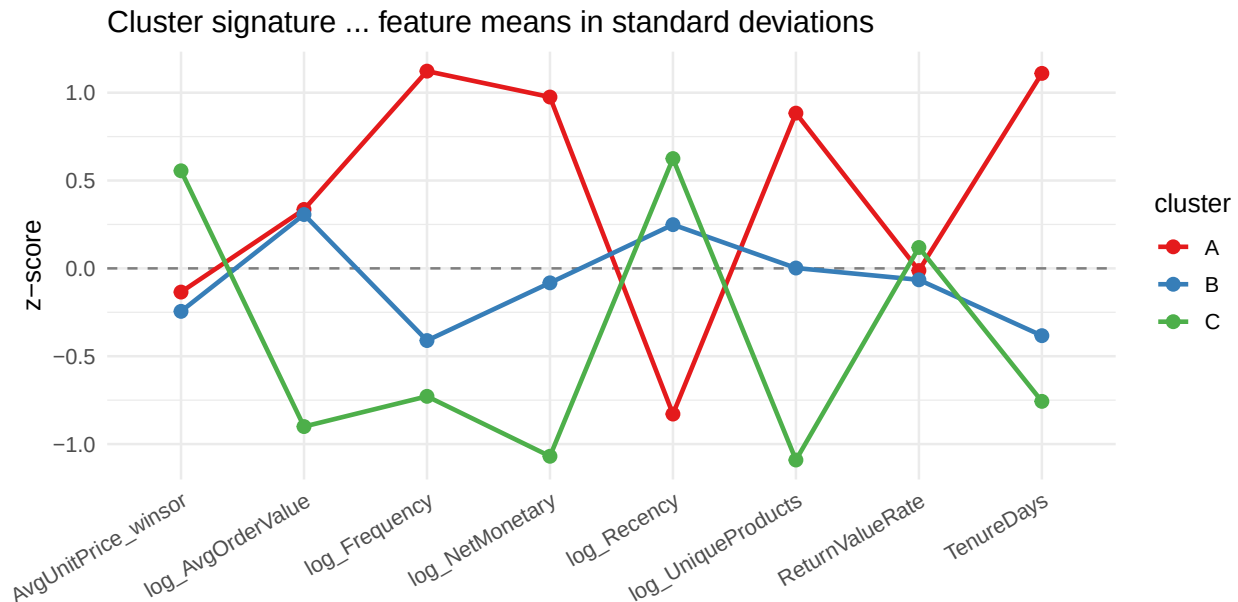
```

z_profile <- cust |>
  select(cluster, log_Recency, log_Frequency, log_NetMonetary,
          log_AvgOrderValue, log_UniqueProducts,
          AvgUnitPrice_winsor, TenureDays, ReturnValueRate) |>
  mutate(across(-cluster, \(x) as.numeric(scale(x)))) |>
  group_by(cluster) |>
  summarise(across(everything(), mean), .groups = "drop") |>
  pivot_longer(-cluster, names_to = "feature", values_to = "z")

ggplot(z_profile, aes(feature, z, group = cluster, colour = cluster)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey50") +

```

```
geom_line(linewidth = 1) +
geom_point(size = 2.5) +
scale_colour_brewer(palette = "Set1") +
labs(title = "Cluster signature - feature means in standard deviations",
      x = NULL, y = "z-score") +
theme(axis.text.x = element_text(angle = 30, hjust = 1))
```

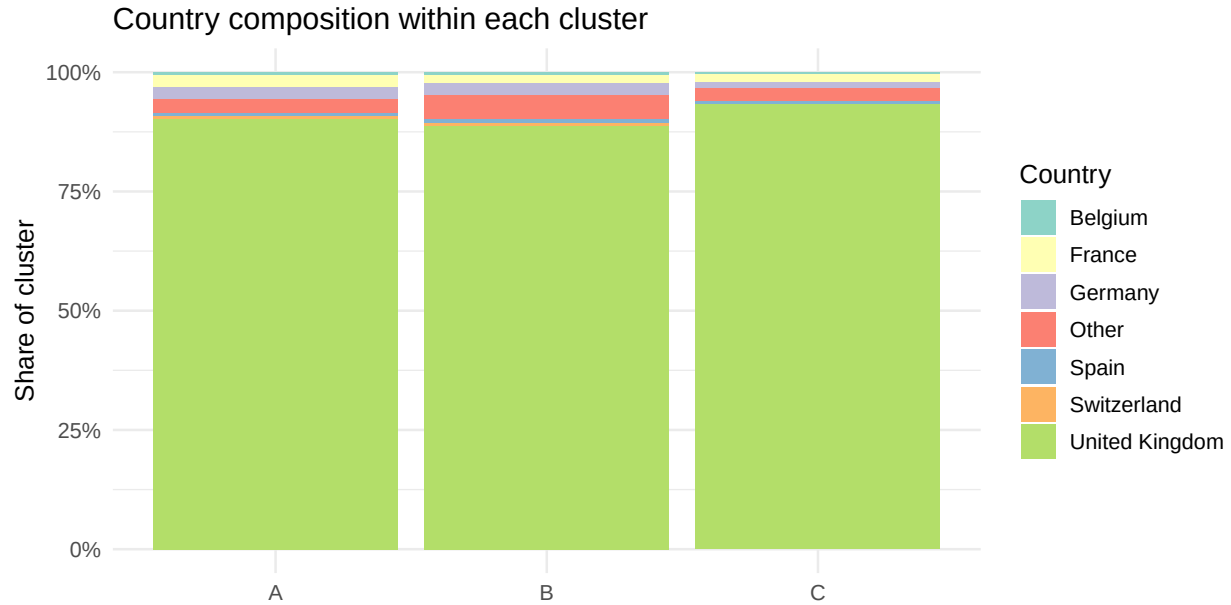


7. Geographic composition

Country was held out of clustering on purpose (UK ~ 83%). Using it now is descriptive only — it tells us whether segments correlate with geography after the fact.

```
top_countries <- cust |> dplyr::count(Country, sort = TRUE) |>
  slice_max(n, n = 6) |> pull(Country)

cust |>
  mutate(Country = ifelse(Country %in% top_countries, Country, "Other")) |>
  dplyr::count(cluster, Country) |>
  group_by(cluster) |>
  mutate(share = n / sum(n)) |>
  ggplot(aes(cluster, share, fill = Country)) +
  geom_col() +
  scale_y_continuous(labels = percent_format()) +
  scale_fill_brewer(palette = "Set3") +
  labs(title = "Country composition within each cluster",
       y = "Share of cluster", x = NULL)
```



8. Top products per cluster — with AND without anomaly

```
anomaly_code <- "23843"

top_with <- tx |>
  group_by(cluster, StockCode, Description) |>
  summarise(revenue = sum(Amount), .groups = "drop") |>
  group_by(cluster) |>
  slice_max(revenue, n = 5) |>
  ungroup()

top_with |>
  mutate(Description = str_trunc(Description, 35),
         revenue      = dollar(revenue, prefix = "pound")) |>
  kable(caption = "Top 5 products by revenue, per cluster (INCLUDING 23843)")
```

Table 3: Top 5 products by revenue, per cluster (INCLUDING 23843)

cluster	StockCode	Description	revenue
A	22423	REGENCY CAKESTAND 3 TIER	£120,435
A	85123A	WHITE HANGING HEART T-LIGHT HOLDER	£75,188
A	85099B	JUMBO BAG RED RETROSPOT	£73,401
A	47566	PARTY BUNTING	£52,392
A	84879	ASSORTED COLOUR BIRD ORNAMENT	£45,368
B	23843	PAPER CRAFT , LITTLE BIRDIE	£168,470
B	23166	MEDIUM CERAMIC TOP STORAGE JAR	£77,942
B	22502	PICNIC BASKET WICKER 60 PIECES	£39,620
B	85123A	WHITE HANGING HEART T-LIGHT HOLDER	£20,844

cluster	StockCode	Description	revenue
B	22423	REGENCY CAKESTAND 3 TIER	£15,386
C	22423	REGENCY CAKESTAND 3 TIER	£6,444
C	85123A	WHITE HANGING HEART T-LIGHT HOLDER	£4,361
C	47566	PARTY BUNTING	£2,928
C	85099B	JUMBO BAG RED RETROSPOT	£2,774
C	48138	DOORMAT UNION FLAG	£2,699

```
top_without <- tx |>
  filter(StockCode != anomaly_code) |>
  group_by(cluster, StockCode, Description) |>
  summarise(revenue = sum(Amount), .groups = "drop") |>
  group_by(cluster) |>
  slice_max(revenue, n = 5) |>
  ungroup()

top_without |>
  mutate(Description = str_trunc(Description, 35),
         revenue      = dollar(revenue, prefix = pound)) |>
  kable(caption = "Top 5 products by revenue, per cluster (EXCLUDING 23843)")
```

Table 4: Top 5 products by revenue, per cluster (EXCLUDING 23843)

cluster	StockCode	Description	revenue
A	22423	REGENCY CAKESTAND 3 TIER	£120,435
A	85123A	WHITE HANGING HEART T-LIGHT HOLDER	£75,188
A	85099B	JUMBO BAG RED RETROSPOT	£73,401
A	47566	PARTY BUNTING	£52,392
A	84879	ASSORTED COLOUR BIRD ORNAMENT	£45,368
B	23166	MEDIUM CERAMIC TOP STORAGE JAR	£77,942
B	22502	PICNIC BASKET WICKER 60 PIECES	£39,620
B	85123A	WHITE HANGING HEART T-LIGHT HOLDER	£20,844
B	22423	REGENCY CAKESTAND 3 TIER	£15,386
B	21108	FAIRY CAKE FLANNEL ASSORTED COLOUR	£14,622
C	22423	REGENCY CAKESTAND 3 TIER	£6,444
C	85123A	WHITE HANGING HEART T-LIGHT HOLDER	£4,361
C	47566	PARTY BUNTING	£2,928
C	85099B	JUMBO BAG RED RETROSPOT	£2,774
C	48138	DOORMAT UNION FLAG	£2,699